

BANARAS HINDU UNIVERSITY

B.Voc. Semester II Examination, 2022-23

Paper: BVCA-201 — Introduction To Databases

Time: Three Hours

Maximum Marks: 70

SECTION — A: Long Answer Questions

[12 × 2 = 24]

Instructions: Answer any two questions. Each question carries 12 marks.

Question 1. Draw an E-R Diagram for a hospital management system in detail and explain it.

Question 2. What is an E-R Diagram? Explain its components.

Question 3. What is normalization? Explain it with different normal forms.

Question 4. What is the difference between a File System and a DBMS? Explain any six applications of Database Management Systems.

SECTION — B: Short Answer Questions

[6 × 6 = 36]

Instructions: Answer any six questions. Each question carries 6 marks.

Question 5. What is a Database and a DBMS? Explain with a suitable example.

Question 6. What is Data Abstraction in DBMS and what are its three levels?

Question 7. What are the different types of SQL commands?

Question 8. What is the difference between TRUNCATE, DELETE, and DROP statements?

Question 9. What are UNION, MINUS, and INTERSECT commands in DBMS?

Question 10. What is a Trigger in DBMS?

Question 11. What are the advantages of a DBMS?

Question 12. What is the difference between a primary key and a foreign key?

Question 13. What is a data model? List different types of data models and explain.

SECTION — C: Multiple Choice Questions

[1 × 10 = 10]

Instructions: Answer all questions. Each question carries 1 mark.

Question 14. Answer the following multiple choice questions:

- (a) What do you mean by a one-to-many relationship?
- (i) One class may have many teachers.
 - (ii) One teacher can have many classes.
 - (iii) Many classes may have many teachers.
 - (iv) Many teachers may have many classes.
- (b) A Database Management System is a type of _____ software.
- (i) System software.
 - (ii) Application software.
 - (iii) General software.
 - (iv) Both (a) and (c).
- (c) Which of the following refers to the level of data abstraction that describes exactly how the data is actually stored?
- (i) Conceptual Level.
 - (ii) Physical Level.
 - (iii) File Level.
 - (iv) Logical Level.
- (d) Which one of the following is a type of Data Manipulation Command?
- (i) Create.
 - (ii) Alter.
 - (iii) Delete.
 - (iv) All of the above.
- (e) The term “Data independence” refers to:
- (i) Data is defined separately and not included in the programs.
 - (ii) Programs are not dependent on the logical attributes of the data.
 - (iii) Programs are not dependent on the physical attributes of the data.
 - (iv) Both (b) and (c).
- (f) What is the relational calculus?
- (i) It is a kind of procedural language.
 - (ii) It is a non-procedural language.
 - (iii) It is a high-level language.
 - (iv) It is a Data Definition language.

- (g) Which one of the following refers to the total view of the database content?
- (i) Conceptual view.
 - (ii) Physical view.
 - (iii) Internal view.
 - (iv) External view.
- (h) The Database Management Query language is generally designed to:
- (i) Support end-users who use English-like commands.
 - (ii) Specifying the structure of the database.
 - (iii) Support in the development of complex application software.
 - (iv) All of the above.
- (i) Which one of the following is commonly used to define the overall design of the database?
- (i) Application program.
 - (ii) Data definition language.
 - (iii) Schema.
 - (iv) Source code.
- (j) Which of the following keys is generally used to represent the relationships between the tables?
- (i) Primary key.
 - (ii) Foreign key.
 - (iii) Secondary key.
 - (iv) None of the above.